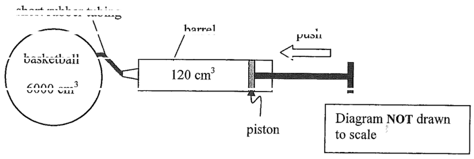
ch7 More about Forces

Q2

*2. Figure 2.1 shows a basketball connected to an air pump via a short rubber tubing. By pushing the piston inward, the pump can compress 120 cm^3 of air inside its barrel at atmospheric pressure and room temperature into the basketball for each stroke.

Figure 2.1



Initially the volume of air inside the basketball is 6000 cm^3 and is at equilibrium with the atmospheric pressure 100 kPa. The basketball has to be pumped to a pressure of 156 kPa for an official match. Throughout the pumping process, the temperature of the basketball and the surroundings is assumed to be maintained at room temperature which is constant.

For each stroke, the valves of the pump (not shown in Figure 2.1) allow the air in the barrel to be completely pumped into the basketball and prevent it from going back into the barrel when the piston is pulled outward.

(a) (i) Show that 3360 cm^3 of air, originally at atmospheric pressure, is required to be pumped into the basketball until its pressure is suitable for an official match. Assume that the volume of the basketball remains unchanged at 6000 cm^3. (3 marks)

(ii) Hence estimate the minimum number of strokes needed to pump the basketball to the required pressure. (1 mark)

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(b) Use kinetic theory of an ideal gas to explain the increase of pressure inside the basketball when air is pumped into it. (2 marks)

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Q5

5. Figure 5.1 shows the use of a beam balance to measure the mass of a load. Without the load and the counter-weight, the beam with a scale pan at its left end is balanced and remains horizontal. ($g = 9.81$ m s^{-2})

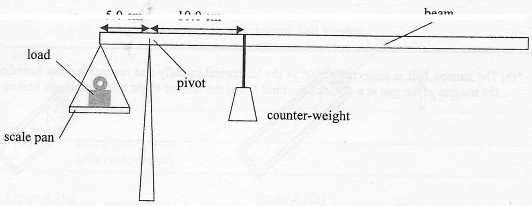

Figure 5.1

(a) A load is placed on the pan which is 5.0 cm from the pivot. The set-up is in equilibrium when the counter-weight of mass 50 g is 10.0 cm away from the pivot as shown.

(i) Find the mass of the load. (2 marks)

(ii) If the reading of the counter-weight's position on the beam carries an uncertainty of ±0.1 cm, find the maximum error that corresponds to the result found in (a)(i). (2 marks)

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(b) The weight of an identical load is now measured by a spring balance calibrated in newtons. What is the balance's reading ? (1 mark)

(c) The beam balance set-up in Figure 5.1 and the spring balance with the load in (b) are both brought into a lift.

(i) The measurements are then repeated inside the lift which accelerates upwards uniformly. State the change, if any, in the measurements respectively. (2 marks)

counter-weight position on beam balance	spring balance reading

(ii) A student claims that if the lift falls freely, the beam balance can still be used to measure the mass of the load. Explain whether the claim is correct or not. (2 marks)

Answers written in the margins will not be marked.